

Ssu-Yuan Lin (Blake)

AI/ML, Software & Algorithm Engineer

ssu.yuan.lin@gmail.com

PROFESSIONAL SUMMARY

Software and algorithm engineer with 3+ years of professional production experience across ML pipelines, computer vision, factory systems, and hardware-software integration. Broader software, research, and personal engineering practice since 2018. Delivered 10x faster model training, 40% lower AOI cycle time, and 66% better 3D measurement accuracy. Current personal engineering applies the same measurement-first discipline to AI workflow reliability, retrieval diagnosis, and verifiable delivery.

CORE SKILLS

Programming: Python, C++, C#, Rust, Kotlin

AI/ML & Vision: TensorFlow, OpenCV, Nvidia Morpheus, PhysicsNeMo, 3D-UNet, stereo vision, Intel RealSense

Systems: ML pipelines, profiling, test design, hardware integration, Docker, Git, GCP, AWS

Agent Engineering: workflow contracts, state machines, deterministic evaluation, mutation testing, recovery, human decision gates

Scientific Computing: FFT, signal analysis, physical modeling, OpenFOAM, Omniverse

Professional Development: Google Grow with Google cloud coursework; weekly AI-paper study group since 2021; system-design practice

Spoken Languages: Mandarin Chinese — Native; English — Professional

PROFESSIONAL EXPERIENCE

Senior AI Engineer — DynaSafe Corp.

Jan 2026 – Apr 2026

- Unblocked model iteration by profiling a dataloader bottleneck, tracing its root cause, and rebuilding the pipeline, improving model-training speed by 10x.
- Built and adapted a Nvidia Morpheus network-anomaly workflow using real PCAP input and PhysicsNeMo training workflows for an OpenFOAM CFD use case.

Software Engineer — MetGauge Co., Ltd.

Sep 2023 – Aug 2024

- Advanced a C++/Python 3D-measurement MVP to factory-ready production testing through system design, hardware integration, and boundary analysis for a 12-stereo-camera line.
- Defined test specifications and analyzed results to guide cross-functional hardware tuning.

Algorithm Engineer — UTECHZONE Corp.

Jun 2021 – Mar 2022

- Developed a C# automated optical inspection algorithm that reduced manufacturing cycle time by 40%.
- Mentored two engineers and worked with a cross-functional team of 10+ on integration milestones.

Software Engineer, Algorithm Development — LIPS Corporation

Jul 2019 – Apr 2020

- Improved real-time stereo-camera 3D measurement accuracy by 66% through calibration and algorithm redesign.
- Calibrated Intel RealSense D415/D435 camera systems.

Earlier research — National Central University, Sep 2018 – Jan 2019: analyzed high-speed microscopy data and developed particle-tracking workflows for bacterial flagellar motor research.

SELECTED PERSONAL ENGINEERING

Study Atlas — Verifiable AI Research Delivery

- Made research-agent delivery auditable: a live Discord request processed a 93,286-character paper, caught a timing violation through eight gates, and produced hash-matching delivery artifacts.

Blake Fleet — Retrieval Failure Diagnosis

- Proved a retrieval failure began with corpus coverage rather than ranking: found 16/19 gold documents missing, expanded the index from 1,281 to 2,710 entries, and improved hit@5 from 0/19 to 6/19.

Agent Ticket Kanban — Enforceable Agent Workflows

- Built compiled work contracts, deterministic state transitions, recovery, and collision-safe writes; a dated verification sweep passed 816 tests and killed 146/146 declared mutations.

EDUCATION

M.S. in Physics, National Central University, 2016–2018

Phi Tau Phi Scholastic Honor Society

B.S. in Physics, National Central University, 2012–2016